NetSy: Network Emissions Synthesizer

- Generalize MASS-GT for different study cases: the same mechanisms, with local evidence and classifications supplied as data.
- Preserve a comparable pipeline: actors, flows, shipments, tours, road loads and emissions.
- Add modelling contributions, including COPERT V emissions informed by Dias & Jenelius (2026).

Comparisons below use MASS-GT's own Dutch study-area data.
Paper: Urban Transitions 5, 100031.

Actors and commodity flows: port + generalization

Indicator	MASS-GT	NetSy
Actors retained	86 004	84 532–85 849
Size retained (jobs)	1 034 015	1 033 511–1 039 856
SIF: max. error after rounding	0.000 tonnes/year	

- Firms become agents of any class; jobs become general stratum-size totals.
- SIF's regressions and FRATAR operate on supplied attributes, resources and regions.

NetSy seeds 1–3; one fixed MASS-GT reference. Ranges show seed sensitivity, not confidence intervals or a formal equivalence test.

Shipments and freight tours: port + generalization

Indicator	MASS-GT	NetSy
Shipments	118 961	118 345–118 637
Daily shipment tonnes	401 438.24	401 438.24
Freight tours	60 062	59 802–60 271
Freight VKT (million)	20.907	20.878–21.091

- Keep sourcing, shipment-size/vehicle choice, delivery windows and end-tour logits.
- Replace Dutch categories with resource/segment strata and typed facilities.

NetSy seeds 1–3; one fixed MASS-GT reference. Ranges show seed sensitivity, not confidence intervals or a formal equivalence test.

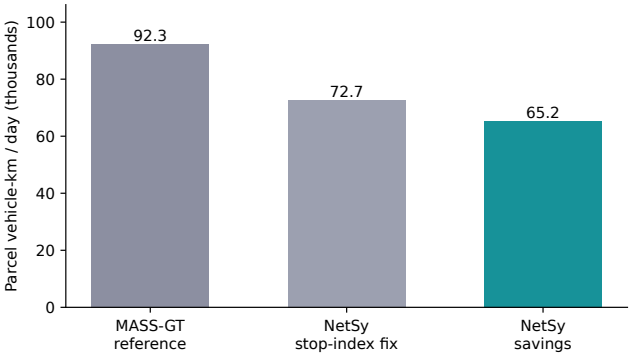
Parcel demand and service vans: port + generalization

Indicator	MASS-GT	NetSy
Customer parcels/day	228 707	228 707
Service van trips/day	367 479.1	367 497.9
Construction van trips/day	178 228.8	178 247.0
Max. OD-cell difference	–	0.008 trips

- Same parcel-demand mechanism; population strata and operators are data.
- Service/construction gravity models share a general matrix-balancing core.

Van comparison uses all OD cells; small numerical differences are retained, not labelled exact matches.

Parcel routes: separate the fix from the contribution



Fix: use the actual next zone when ordering stops.

Contribution: Clarke–Wright savings and stop relocation.

Savings vs. the corrected port: **10.2% less VKT**, with +3.8% tours. Same 228 707 customer parcels.

Same demand, capacity and Dutch skims; VKT from the zone-to-zone skim, including returns.

Assignment and factors: controlled comparison

Indicator	MASS-GT	NetSy
Network links compared	158 806	
Max. connector trip error	–	8.467
Max. non-connector trip error	–	0.000020
CO_2 (tonnes/day)	16 792.874	16 795.387
CO_2 change explained by fix	2.513 tonnes/day	

- Supply MASS-GT's original trips to both assignments: isolate this stage from synthesis randomness.
- Fix urban parcel emissions being overwritten instead of accumulated.

CO_2 residual after accounting for the fix: 0.248 g. Small load residuals occur on connectors (ties and the reference's eight-neighbour link lookup); see the audit.

UCC scenario: port, fixes and hub pooling

Indicator	MASS-GT UCC	NetSy UCC + pooling
Freight tours	62 931	62 709
Parcel tours	1 760	1 505
Customer parcels	228 707	228 707
Network VKT (million)	44.078	43.925
CO_2 (tonnes/day)	16 761.0	16 712.0

Explicit facility legs and carriers; vehicle/combustion choices; electric and hybrid operations. NetSy additionally pools outbound parcels at each hub.

Vehicle shares are counted once per vehicle, with an explicit fuel fallback. Midnight matrix generation and the reference's single-worker hybrid dispatch are repaired.

Independent stochastic runs, documented fixes and explicit outbound hub pooling; this is a scenario comparison, not an exact-match claim. Reference repaired only for execution/midnight handling; behavioural bugs are listed in the audit.

Microhubs: preserve the model, complete its operations

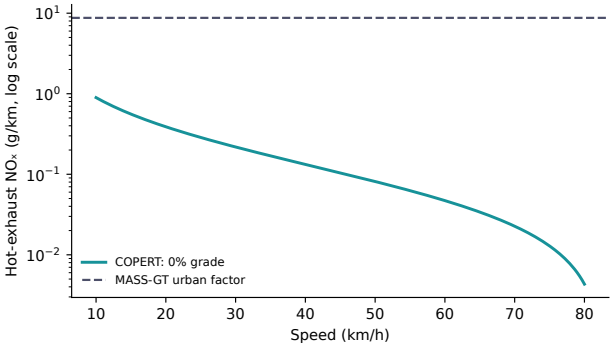
Indicator	MASS-GT (corrected)	NetSy
Hub-bound customer parcels	7 208	7 208
Max. hub–recipient pair error	–	0
Customer parcels delivered	228 707	
Completed parcel tours	No hub-aware schedule	1 811

Port + fixes: hub selection and two parcel legs; correct selected-hub IDs and zone indexing.

Downstream extension: hub-based collaborative tours with vehicle-specific capacity, speed and powertrain.

Collaborative electric-bicycle configuration from the supplied Dutch microhub tables. Raw-reference maximum hub-pair difference: 391 parcels; corrected difference: zero. Both references are retained; zero pair error is against corrected selection. NetSy adds the missing hub-aware downstream operations.

COPERT V: an additional emissions model



Speed, grade and payload select hot-exhaust factors. Keep payload bins separate before aggregation.

Full pipeline, same seed-1 trips: factors: 16 768.2 t CO_2; COPERT: 18 140.3 t CO_2.

A model/fleet sensitivity, not a measured emissions improvement.

Declared Euro VI truck / Euro 6 van fleet; EMEP/EEA Appendix 4 (July 2026). Curve: medium truck, 0% grade, 50% payload. Dutch network run assumes flat grades; no non-exhaust estimate is claimed. Paper informs the speed/grade/payload and hub/powertrain extension.

Implementation improvements and validation boundary

- Array bookkeeping, cached proximity and compiled stop improvement; preserve the baseline tour result.
- Parquet outputs and column-based browser tables. The 6 089 837-row loads file loaded in 2.1 s in the recorded browser check.
- Stage comparisons separate inherited behaviour, generalization, fixes and modelling additions; three NetSy seeds assess stochastic sensitivity, and original-trip controls isolate assignment/emissions.

Scope: the standard Dutch pipeline, UCC and collaborative microhub case. Optional crowdshipping and selected-link analyses are not benchmarked here. Benchmark agreement is distinct from empirical validity of a new study case. Code, patches, inputs and comparison scripts are identified in validation/.